

We define $f(x, y)$ as no. of different corresponding bits in the binary representation of x and y .

For example $f(2, 7) = 2$, since the binary representation of 2 and 7 are 010 and 111, respectively. The first 2 and the 3rd bit differ, so $f(2, 7) = 2$.

You are given an array of n positive integers $A_1, A_2, \dots$. Find the sum of $f(A_i, A_j)$ for all pairs (i, j) such that $1 \leq i, j \leq n$. Return the answer modulo $10^6 + 7$.

Input 1:

$A = [1, 3, 5]$

Output:

~~8~~ 8

Input 2:

$A = [2, 3]$

2

Algorithm:

for each bit position

(1) count how many no. have bit = 1
call it count 1

(2) Remaining no. have bit = 0
count = $n - \text{count 1}$

Now every pair where one no. has that bit 1 and one with 0 contributes 1 to answer.

No. of such pairs = $\text{count 1} \times \text{count 0}$

for (bit = 0 to bit = 32)
count1 = 0

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for (i = 0 to n) {
    if (num[i] & (1 << bit))
        count1++;
}
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count0 = n - count1
ans += count1 * count0
return ans * 2

Time Complexity:

$$O(32 \times n) = O(n)$$

$$\text{Formula} = \sum_{i=0}^{31} (\text{count0} \times \text{count1})$$